

Which retriever finds the right rule *when a citizen asks about a welfare scheme?*

A controlled evaluation of five retrieval strategies on 22 Indian government policy documents.

22

POLICY DOCUMENTS

34

HAND-GRADED QUESTIONS

183

HUMAN RELEVANCE JUDGEMENTS

5

RETRIEVAL SYSTEMS

Somebody asks whether they qualify. The system answers confidently.

Government departments, NGOs and citizen helpdesks are deploying RAG assistants over scheme guidelines. The failure that matters is not a clumsy sentence — it is a confident answer built on the wrong document.

THE QUESTION

"I earn ₹18,000 a month and live in the city. Can I get help buying a house?"

WHAT SHOULD BE RETRIEVED

The PMAY-Urban guideline defining EWS, LIG and MIG income bands and the benefit each receives.

WHAT GOES WRONG

A retriever returns the rural housing scheme instead. The generator answers faithfully — from the wrong rulebook.

The generator cannot repair evidence it never received. Retrieval decides whether the answer is safe.

Policy language does not look like the text these retrievers were trained on.

Embedding models are trained on web and encyclopaedic text. Indian scheme documentation is a different register: dense with acronyms, statutory phrasing, and administratively defined categories that carry precise legal meaning.

PMAY-G	Pradhan Mantri Awas Yojana – Gramin	<i>rural housing</i>
EWS / LIG / MIG	Economically Weaker Section, Low and Middle Income Group	<i>statutory income bands</i>
PM JANMAN	PM Janjati Adivasi Nyaya Maha Abhiyan	<i>Particularly Vulnerable Tribal Groups</i>
PMJDY	Pradhan Mantri Jan Dhan Yojana	<i>financial inclusion</i>

An embedding trained on Wikipedia has no reason to place "MIG" near "middle income household" — but a keyword index matches it exactly, every time.

The field measured this on web search, then generalised the conclusion.

WHAT THE BENCHMARKS MEASURE

MS MARCO • 500,000 queries
Natural Questions • 300,000 queries

Open-domain web text. Large, diverse vocabulary. Heavy paraphrase between question and passage. On this material dense embeddings reliably beat lexical baselines — and that finding became the default assumption.

WHAT A POLICY CORPUS ACTUALLY IS

22 documents • bounded vocabulary

Scheme names repeat verbatim. Terminology is administratively fixed. Paraphrase is limited because the legal wording is the wording. None of the properties that make dense retrieval win are present — and nobody had published a controlled measurement on this material.

So the honest position before this work was: nobody actually knew. The assumption had never been tested here.

That is the gap this dissertation exists to close.

Research question

On a domain-specific government policy corpus, which retrieval strategy actually puts the correct evidence in front of the generator — and at what operational cost?

Answerable only by measurement

A system demonstration proves nothing. The question requires a benchmark with human relevance judgements and systems frozen before evaluation.

Answerable only on this domain

Results from MS MARCO cannot be borrowed. The corpus properties that drive the answer are the ones being studied.

Answerable only with cost attached

A method that wins by 0.02 MRR and costs 20 seconds per query is not a win for a citizen-facing service.

No benchmark existed, so building one was part of the contribution.

There is no public retrieval benchmark over Indian scheme documentation with human relevance judgements. Constructing one — and doing it defensibly — was a prerequisite to answering anything.

22

SOURCE DOCUMENTS

PMAY-G, PMAY-U, PMJDY, Ujjwala, PM-KISAN, PMFBY, PMKVY, PM JANMAN, PM SVANidhi and others — all public domain

34

QUESTIONS

Six categories: exact lookup, terminology, paraphrase, entity relation, multi-hop, synthesis

183

RELEVANCE JUDGEMENTS

Pooled across all five systems so no system is graded on its own candidate set

0.7421

INTER-RATER K

114-row blind regrade; 14 disagreements adjudicated by hand

Every system saw identical chunks and identical judgements. All configurations were frozen before evaluation.

Five strategies. One corpus. One set of judgements.

BM25

LEXICAL

Probabilistic keyword ranking, 1994. No training, no embeddings.

FAISS

DENSE

Sentence-transformer embeddings, windowed-max scoring over chunks.

Entity Graph

STRUCTURED

Entity co-occurrence graph, 2-hop traversal over extracted entities.

Hybrid RRF

FUSION

Reciprocal rank fusion of BM25 and graph rankings.

Prompt-RAG

LLM RERANK

Claude Haiku reranking the top 50 lexical candidates.

IDENTICAL INPUTS

Same 140 chunks, same chunk IDs, same 34 questions for every system.

POOLED JUDGEMENTS

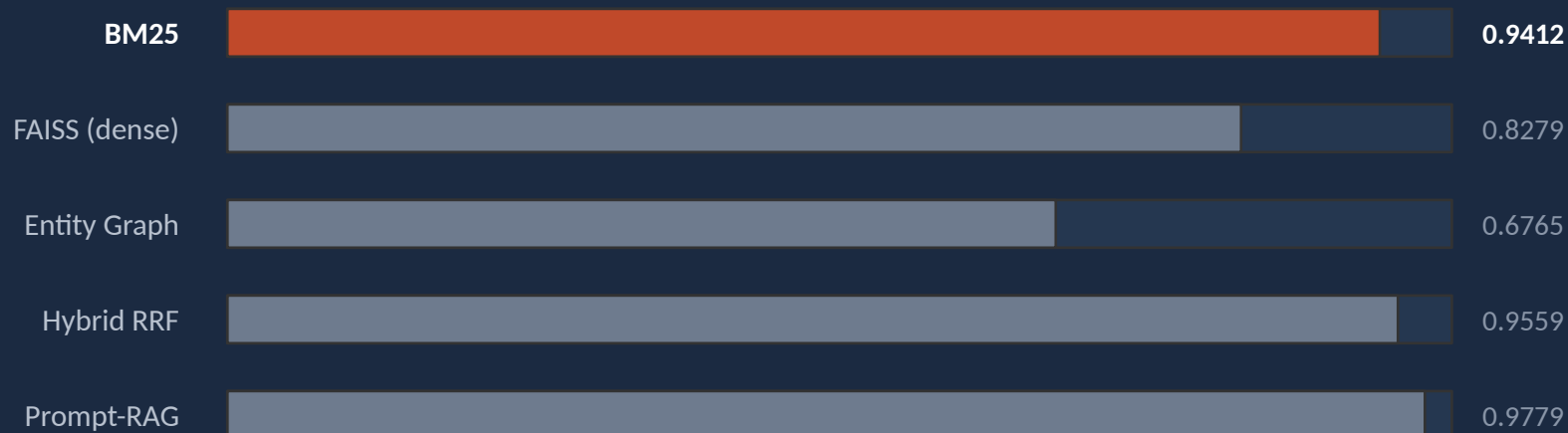
Relevance graded over the union of all systems' candidates, blind to system identity.

FROZEN BEFORE SCORING

Configurations, hashes and manifests fixed in advance. Zero retries; failed runs preserved, not replaced.

A keyword algorithm from 1994 beat modern dense embeddings *on every question category but one.*

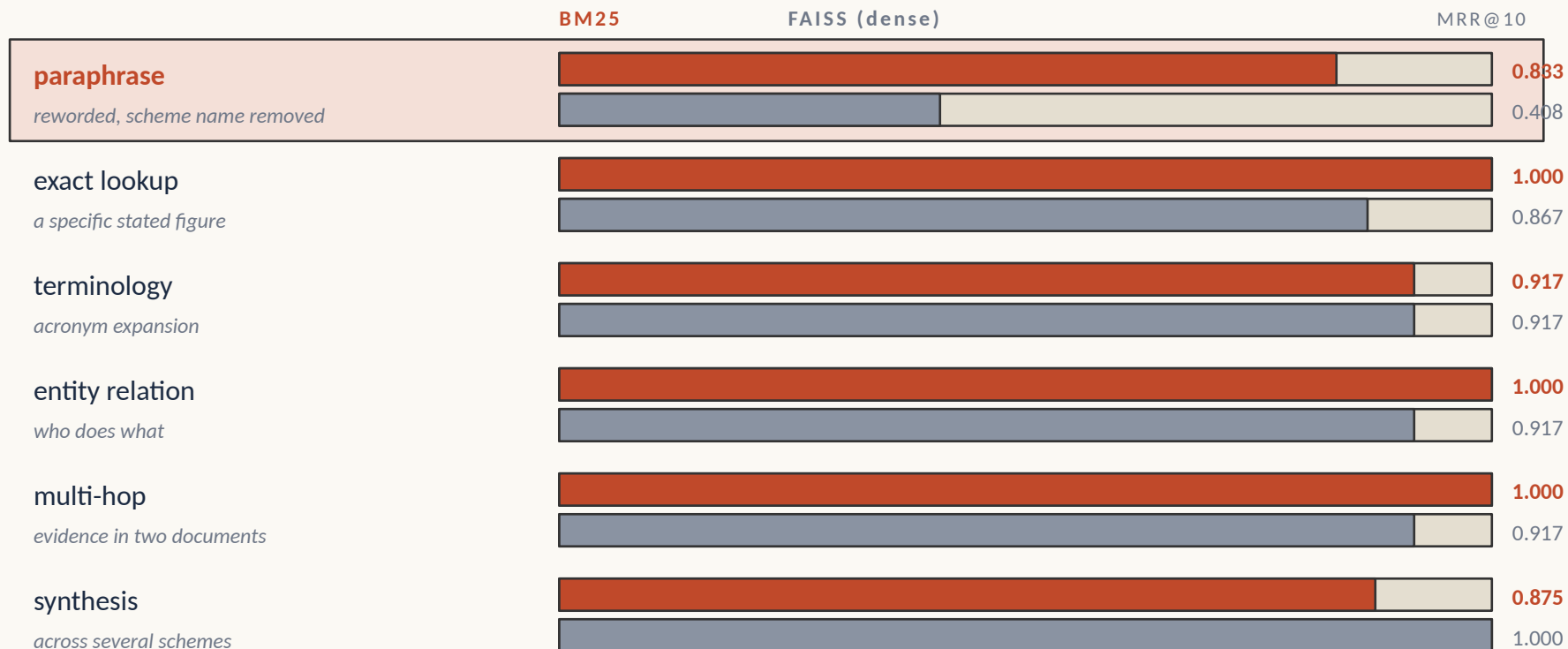
AGGREGATE MRR@10 · 34 HUMAN-VALIDATED QUESTIONS



**BM25 is also the
cheapest system in
the study.**

0.76 ms mean query
latency, against 13.2 ms
for FAISS and roughly 18
seconds for LLM
reranking. It needs no
GPU, no embedding
model and no API
budget.

The gap is widest exactly where dense retrieval was supposed to win.



THE PARAPHRASE RESULT

H2 predicted dense retrieval would win here.

It lost by half. These are questions deliberately rewritten to remove the scheme name — "smoke-producing solid cooking materials" instead of Ujjwala. The case embeddings exist for.

The entity graph scored 0.000: with no scheme name in the query, its linker had nothing to anchor to.

What that actually looks like.

QUESTION v2q-018 • PARAPHRASE

"Which initiative helps city households across officially defined lower and middle income bands obtain permanent homes?"

CORRECT ANSWER • Pradhan Mantri Awas Yojana - Urban (EWS, LIG, MIG households)

BM25

rank 1

correct passage first

FAISS

rank 5

four wrong schemes above it

Entity Graph

not in top 10

no scheme name to anchor to

Hybrid RRF

rank 1

correct passage first

Prompt-RAG

rank 1

correct passage first

The question shares no scheme name with the document. BM25 still anchored on "income", "households" and "homes"; the embedding spread its probability mass across thematically similar but incorrect schemes.

► Live demo — all 34 questions, all five systems

Why lexical matching wins on this corpus.

This is not a claim that BM25 is better than dense retrieval. It is a claim about a corpus with specific statistical properties — and the mechanism is identifiable.

01 Vocabulary is bounded and repeated

A scheme name is effectively a primary key. It appears verbatim in the guideline, the FAQ and the amendment. Lexical matching exploits that identity directly.

02 Rare terms carry the signal

BM25 weights rare terms heavily. In this corpus the rare terms are precisely the discriminative ones — scheme acronyms, statutory categories, named ministries.

03 Embeddings dilute what matters

A general-purpose encoder compresses a passage into one vector, spreading weight across thematically related schemes. Housing documents look alike in embedding space even when only one is correct.

04 Entity extraction is the graph's ceiling

Graph retrieval inherits every NER failure. On paraphrased questions with no named entity, it has no entry point at all — hence 0.000, not a tuning problem.

The best-scoring system is not the one you would deploy.

SYSTEM	MRR@10	MEAN LATENCY	API COST	INFRASTRUCTURE
BM25	0.9412	0.76 ms	none	CPU only
FAISS	0.8279	13.2 ms	none	embedding model + index
Entity Graph	0.6765	0.56 ms	none	NER pipeline
Hybrid RRF	0.9559	0.93 ms	none	BM25 + graph
Prompt-RAG	0.9779	≈18 s	per query	LLM API

Prompt-RAG buys 0.037 MRR over BM25 for roughly 24,000× the latency and a per-query API charge. For a helpdesk answering thousands of citizen queries a day, that is not a trade a deployment can justify — and H4 is better read as a Pareto frontier than a winner.

Four of five predictions were wrong. That is the result, not a shortfall.

Hypotheses were registered before evaluation and scored against a frozen protocol: paired whole-query bootstrap, exact sign-randomisation, Holm correction across 32 tests. Nothing was reinterpreted after seeing the data.

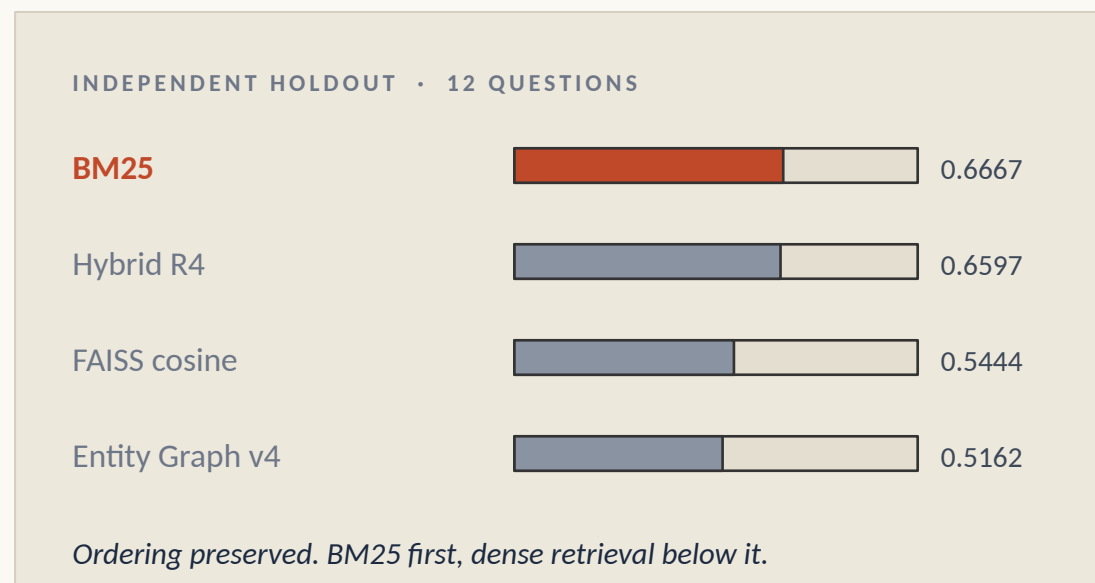
H1	BM25 competitive with FAISS on exact and terminology queries	Inconclusive	Equivalence needs the interval inside ± 0.05 . With 12 questions it is not established either way.
H2	FAISS beats BM25 on paraphrase	Not supported	Reversed: BM25 0.833 vs 0.408. Direction repeats on holdout. The informative failure.
H3	Graph beats both on entity and multi-hop	Not supported	Pilot rejects it outright. Holdout shows the strongest directional signal of any hypothesis — unresolved.
H4	Hybrid achieves highest aggregate MRR	Not supported	Hybrid 0.9559 vs Prompt-RAG 0.9779. Reframed as a cost-quality frontier.
H5	Faithfulness independent of retrieval quality	Partially supported	Faithfulness is enforced at generation; retrieval sets what can be said, not how faithfully.

A preregistered study that confirms everything it predicted has usually not been preregistered.

Does it hold outside the pilot?

Twelve fresh questions on an independently acquired document set, with every configuration frozen beforehand.

Separately, the corpus was scaled to 130 documents to test whether the pipeline holds.



SCALE TEST • 130 DOCUMENTS, 100 QUESTIONS	
HELD	FAISS remained weakest or near-weakest. BM25, Hybrid and Prompt-RAG each beat it on nDCG after Holm correction.
CHANGED	Prompt-RAG led the pilot on nDCG; at scale BM25 led by 0.0004 — a practical tie. Benchmark construction moved the ordering.
NOT CLAIMED	R4 labels are AI-assigned with no owner-labelled overlap. It demonstrates engineering scalability, not confirmatory validity.

All main paired confidence intervals cross zero. These are directional findings that reproduce, not decisive superiority claims.

What exists now that did not before.

A reusable benchmark

22 policy documents, 34 categorised questions, 183 pooled human relevance judgements at $\kappa = 0.7421$. Released with hashes and manifests so any retriever can be scored against it.

A measured counter-result

On domain-specific policy text, lexical retrieval matches or beats dense embeddings — with the mechanism identified and the failure mode isolated to a specific query type.

A cost-aware recommendation

For a citizen-facing policy assistant, BM25 or BM25-plus-fusion is the defensible default. LLM reranking must justify its latency against a measured gain, not an assumed one.

A fully auditable pipeline

Frozen configurations, zero retries, preserved failures, per-query traces. Every number in this talk can be recomputed from the repository in one command.

The benchmark is the part that outlives the result. Anyone can now disagree with me and prove it.

What I am not claiming.

Sample size

34 pilot and 12 holdout questions. Intervals are wide; I report direction and mechanism, not decisive superiority.

Construct validity of paraphrase

Some rewritten questions retained document terminology, weakening the intended lexical-mismatch test. A paraphrase-only follow-up is preregistered.

Scale evidence is engineering

The 130-document run used AI-assigned labels with no human overlap. It shows the pipeline holds, not that the findings confirm.

Single language, single domain

English-language Indian policy text. No claim transfers to other domains or languages without re-measurement.

Thank you · Questions

Every figure recomputes from the repository · [runs/canonical_v2_pilot/](#)